

M – Tech Production & Marketing

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PROJECT WEEK 3

AI Facial Emotion Detector

Introduction

This report summarizes the activities completed during the **M-Tech Production & Marketing AI/ML Internship Program**. Today's session focused on the design and development of the **AI Facial Emotion Detector**, a desktop application capable of recognizing human facial emotions from uploaded images using Artificial Intelligence and Computer Vision techniques.

The project integrates image processing, facial analysis, and an interactive Python-based graphical user interface developed using **Tkinter** and **ttkbootstrap**. The application analyzes facial expressions from uploaded images and predicts emotions such as Happy, Sad, Angry, Fear, Surprise, Neutral, and Disgust. The detected emotion is displayed along with its confidence score, providing users with an intelligent and user-friendly emotion recognition system.

Objectives

- Successfully develop the AI Facial Emotion Detector desktop application.
- Implement an Artificial Intelligence model for facial emotion recognition.
- Design a professional graphical user interface using Python Tkinter and ttkbootstrap.
- Integrate image processing and facial emotion analysis into the application.
- Display detected emotion along with prediction confidence.

- Maintain prediction history for multiple analyses.
- Test and verify the complete application functionality.

Task Description

The assigned task was to develop a fully functional **AI Facial Emotion Detector** desktop application capable of recognizing human emotions from facial images.

The application allows users to upload facial images through an interactive graphical interface. After image preprocessing and facial analysis, the Artificial Intelligence model predicts the dominant facial emotion and displays the prediction with its confidence percentage.

The project also includes additional features such as prediction history, image preview, clear image functionality, status indicators, and a modern graphical interface designed for an improved user experience.

Work Done

Today, I successfully developed the **AI Facial Emotion Detector** by integrating image upload functionality, graphical user interface components, and Artificial Intelligence-based emotion analysis into a single desktop application.

I designed a professional interface using **Tkinter** and **ttkbootstrap**, allowing users to upload facial images and receive emotion predictions in real time. The system processes the uploaded image, analyzes facial expressions, and displays the detected emotion together with its confidence percentage.

Additionally, I implemented image preview functionality, prediction history tracking, status updates, clear image functionality, and improved the overall application layout to provide a clean and professional user experience.

After completing the implementation, I performed multiple tests using different facial images to verify prediction accuracy, interface responsiveness, and overall application performance. Finally, I organized the project files, source code, screenshots, and documentation for professional submission.

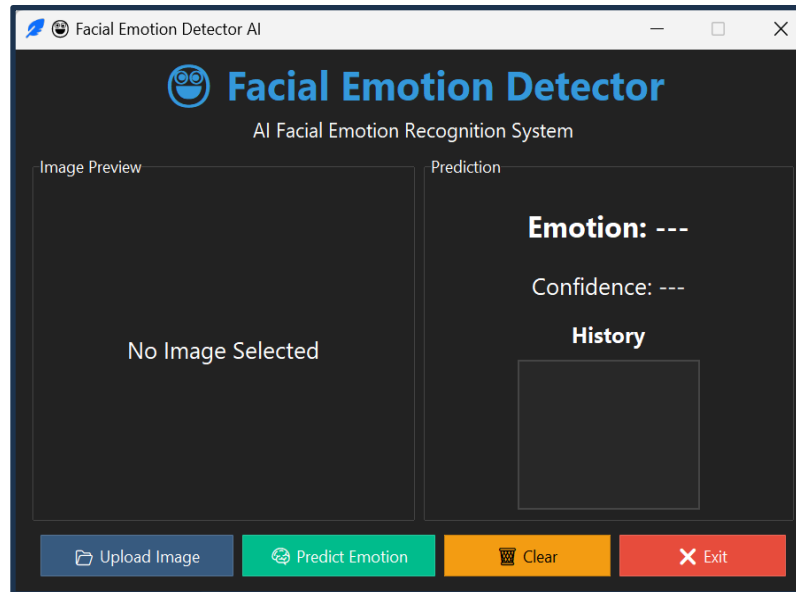


Figure 1: Main Graphical User Interface (GUI)

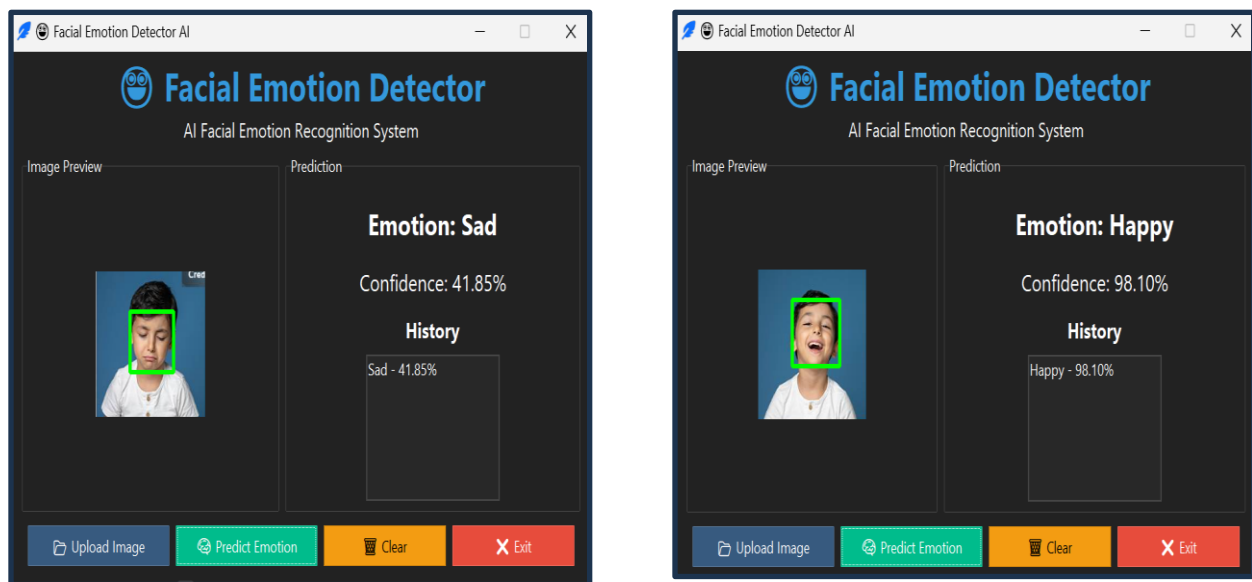


Figure 2: Uploaded Image and Emotion Prediction

Challenges Faced

During the development of the project, several challenges were encountered while integrating the Artificial Intelligence emotion detection model, managing Python package dependencies, processing uploaded images, and improving the graphical user interface.

Additional challenges included handling image preprocessing, ensuring smooth interaction between the AI model and the GUI, and resolving compatibility issues related to AI libraries and TensorFlow.

These issues were resolved through debugging, dependency management, application testing, and continuous improvements to the system design until all major functionalities worked successfully.

Key Learnings

Today's project significantly enhanced my practical understanding of **Artificial Intelligence, Computer Vision, and Desktop Application Development**.

I learned how to preprocess facial images, integrate AI-based emotion recognition models into Python applications, develop professional graphical user interfaces using Tkinter, manage prediction history, handle user interactions, and troubleshoot dependency-related issues.

This project also strengthened my debugging skills, problem-solving abilities, and practical knowledge of developing intelligent real-world AI applications.

Conclusion

The **AI Facial Emotion Detector** project was successfully developed by integrating Artificial Intelligence, image processing, and graphical user interface design into a single desktop application.

The application efficiently analyzes uploaded facial images and predicts human emotions while providing confidence scores and an intuitive user interface. Completing this project further strengthened my practical experience in Computer Vision, Artificial Intelligence, GUI development, and real-world software implementation, bringing me another step closer to becoming a skilled AI Engineer.

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